Cover inequalities

Outline

- Why cover inequalities?
- A constraint with positive coefficients and its LP relaxation
- Covers and cover inequalities
  - Minimal covers
  - The cover closure
- Geometric interpretation
- Separation of cover inequalities
  - An ILP formulation for the CSP
- Transforming the CSP into a complementary problem
  - A greedy separation heuristic
- Extended cover inequalities
- Violation and distance from a cover inequality
- Typical mistakes
- In summary

Learning objectives

At the end of this chapter you will be able to:

- write the cover inequalities of a $\leq$ constraint with positive coefficients of a binary MIP, and prove that they are valid;
- list the covers and the minimal covers of an instance, and describe the cover closure;
- separate a cover inequality from a fractional point by solving a cover separation problem, exactly or with the greedy heuristic algorithm;
- strengthen a cover inequality into an extended cover inequality;
- compute the violation of a cover inequality and the distance of a point from it.

Prerequisites

From this module:

- Chapter 1 «Integer linear programming» and Chapter 2 «Primal and dual bounds»: ILP problems, the LP relaxation, and the upper bound given by the LP relaxation
- Chapter 3 «Ideal formulations» and Chapter 4 «Valid inequalities»: the convex hull of the integer solutions, valid inequalities, and cuts that remove fractional points
- The LP relaxation of a binary problem with one constraint, solved by inserting the variables in non-increasing order of the ratio c_j/a_j between objective and constraint coefficients (Dantzig's rule)

Why cover inequalities?

Introduction: Why cover inequalities?

- Many binary MIPs contain a **knapsack** (or **budget**) constraint: a $\leq$ constraint with positive coefficients
- The LP relaxation of such a constraint often has fractional optimal solutions, and gives a weak upper bound
- We look for valid inequalities that are easy to write and to find, and that cut off these fractional points

From valid inequalities to cover inequalities

- In the chapter «Valid inequalities» we added **valid inequalities** to tighten the LP relaxation: each one removes some fractional points and keeps all the integer ones
- For a knapsack constraint there is a whole family of such inequalities, built from the combinatorial structure of the constraint: the **cover inequalities**

A challenge

- Consider the instance with four binary variables, right-hand side $b = 9$, and the following objective coefficients c_j and constraint coefficients a_j:

j	1	2	3	4
c_j	6	5	8	9
a_j	2	3	6	7

The optimal solution of its LP relaxation is **fractional**.

Can you find an inequality that cuts it off
without removing any feasible binary point?

- The examples of this chapter answer this question

Summary

- Knapsack constraints with positive coefficients appear in many binary MIPs
- Their LP relaxation can have **fractional** optimal solutions
- **Cover inequalities** cut off such points and keep every feasible binary point
- The **four-variable instance** will follow us through the chapter

A constraint with positive coefficients and its LP relaxation

Introduction: A constraint with positive coefficients

- We define the object of the chapter: one $\leq$ constraint with positive integer coefficients on binary variables
- We define its set X of feasible binary points and its LP relaxation P
- The goal is to strengthen P: remove fractional points without removing any point of X

Definition 1: constraint with positive coefficients of a binary MIP (1/2)

Given binary variables x_j, for $j \in \mathscr{I} = \{1, 2, \ldots, n\}$, integer coefficients $a_j \in \mathbb{Z}_{\geq 1}$ and a right-hand side $b \in \mathbb{Z}_{\geq 1}$, consider the constraint

$$\sum_{j \in \mathscr{I}} a_j x_j \leq b,$$

$$x_j \in \{0, 1\}, \quad \forall j \in \mathscr{I}.$$

Definition: constraint with positive coefficients of a binary MIP (2/2)

Its set of feasible solutions is

$$X = \left\{ \boldsymbol{x} \in \{0,1\}^n : \sum_{j \in \mathscr{J}} a_j x_j \leq b \right\},$$

and its LP relaxation is

$$P = \left\{ \boldsymbol{x} \in [0,1]^n : \sum_{j \in \mathscr{J}} a_j x_j \leq b \right\}.$$

Knapsack constraints

- Constraints of this type are called **knapsack constraints** or **budget constraints**
- We assume that

$$a_j \leq b \quad \text{for all} \quad j \in \mathscr{I} \qquad \text{and} \qquad \sum_{j \in \mathscr{I}} a_j > b$$

- The first assumption guarantees that every variable can be set to 1
- The second one guarantees that they cannot all be set to 1 together

Strengthening the LP relaxation

- X is a finite set of binary points, and an exact description of its convex hull $\text{conv}(X)$ is generally hard to obtain
- The aim is to strengthen P by removing fractional points without removing any point of X
- A MIP can contain several constraints of this type, and further constraints
- Valid inequalities for X remain valid when other constraints are imposed

An upper bound from P

With objective coefficients c_j, maximizing $\sum_{j \in \mathscr{J}} c_j x_j$ over P gives an **upper bound** on the integer maximization problem.

Valid inequalities for X
can improve this upper bound.

Summary

- One constraint $\sum_{j \in \mathscr{J}} a_j x_j \leq b$ with $a_j \in \mathbb{Z}_{\geq 1}$, $b \in \mathbb{Z}_{\geq 1}$ and binary variables: a **knapsack constraint**
- X: its feasible binary points; P: its LP relaxation in $[0,1]^n$
- Assumptions: $a_j \leq b$ for all j, and $\sum_{j \in \mathscr{J}} a_j > b$
- Valid inequalities for X strengthen P and can improve the upper bound

Covers and cover inequalities

Introduction: Covers and cover inequalities

- A **cover** is a set of variables that cannot all be equal to 1 together
- Each cover gives a valid inequality: the **cover inequality**
- We see that the minimal covers are enough, define the cover closure, and compute it on the four-variable instance

Covers

Definition 2: Cover and collection of all covers

A cover is a subset $C \subseteq \mathscr{I}$ whose total coefficient $\sum_{j \in C} a_j$ exceeds the right-hand side b. The collection of all covers is

$$\mathcal{C} = \left\{ C \subseteq \mathscr{I} : \sum_{j \in C} a_j > b \right\}.$$

Since coefficients and right-hand side are integer, the condition is equivalent to $\sum_{j \in C} a_j \geq b + 1$.

Cover inequalities

Definition 3: Cover inequality

Given $C \in \mathcal{C}$, the associated cover inequality is

$$\sum_{j \in C} x_j \leq |C| - 1.$$

- It allows at most $|C| - 1$ of the variables of C to be equal to 1

Validity of cover inequalities

Proposition 1

For every cover $C \in \mathcal{C}$, the cover inequality

$$\sum_{j \in C} x_j \leq |C| - 1$$

is valid for X.

Proof

- Suppose that a point $\boldsymbol{x} \in X$ violates the cover inequality
- The sum of $|C|$ binary variables is then greater than $|C| - 1$, hence $x_j = 1$ for every $j \in C$
- Since all coefficients are positive,

$$\sum_{j \in \mathscr{J}} a_j x_j \geq \sum_{j \in C} a_j > b$$

- This contradicts the constraint $\sum_{j \in \mathscr{J}} a_j x_j \leq b$: at most $|C| - 1$ variables of C can be equal to 1 together



A cover can contain a smaller cover.

We now see that the **smallest** covers are enough.

Minimal covers

Definition 4: Minimal cover

A cover C is minimal if no proper subset of C is a cover. Equivalently,

$$\sum_{j \in C} a_j > b, \qquad \sum_{j \in C} a_j - a_i \leq b \quad \forall i \in C.$$

- Since the coefficients are positive, it suffices to remove one variable at a time: if each of these $|C|$ subsets satisfies the constraint, every smaller subset satisfies it as well

The minimal covers are enough

Proposition 2

The inequalities of the minimal covers, together with the upper bounds $x_j \leq 1$ for all $j \in \mathscr{J}$, imply every cover inequality.

Proof

- Start from a cover C and remove variables as long as the remaining set is still a cover: this gives a minimal cover $D \subseteq C$
- Add the cover inequality of D and the upper bounds of the variables in $C \setminus D$:

$$\sum_{j \in C} x_j = \sum_{j \in D} x_j + \sum_{j \in C \setminus D} x_j$$

$$\leq |D| - 1 + |C \setminus D|$$

$$= |C| - 1.$$

□

What do we obtain if we add
all the cover inequalities to P?

The cover closure

Definition 5: Cover closure

The cover closure of P is the polyhedron

$$P^{\text{cov}} = \left\{ \boldsymbol{x} \in P : \sum_{j \in C} x_j \leq |C| - 1 \quad \forall C \in \mathcal{C} \right\}.$$

Between $\text{conv}(X)$ and P

- Since all cover inequalities are valid,

$$\text{conv}(X) \subseteq P^{\text{COV}} \subseteq P,$$

  and in general the first inclusion is strict

- By the previous proposition, the inequalities of the minimal covers are sufficient to describe P^{COV}

How many covers?

- The number of covers can be exponential in n
- If $a_j = 1$ for all $j \in \mathscr{I}$ and $b = 1$, every subset with at least two indices is a cover, i.e., there are $2^n - n - 1$ covers
- Also the minimal covers can be exponentially many: for $a_j = 1$ for all $j \in \mathscr{I}$, $n = 2m$ and $b = m$, they are the $\binom{2m}{m+1}$ subsets of $m + 1$ indices

Adding only what is needed

Cover inequalities are **not** added all at once:
they are added only when they are violated,

i.e., they are **separated**.

- The separation of cover inequalities is the subject of a later section of this chapter

Example: a binary problem with one constraint (1/3)

- Consider $\mathscr{I} = \{1, 2, 3, 4\}$, right-hand side $b = 9$, and the following objective coefficients c_j and constraint coefficients a_j:

j	1	2	3	4
c_j	6	5	8	9
a_j	2	3	6	7

- This is the **four-variable instance**, the guiding example of the chapter

Example: a binary problem with one constraint (2/3)

- The associated LP model is:

$$
\begin{array}{rrcrcrcrcl}
\max & 6\,x_1 & + & 5\,x_2 & + & 8\,x_3 & + & 9\,x_4 & & \\
\text{subject to} & 2\,x_1 & + & 3\,x_2 & + & 6\,x_3 & + & 7\,x_4 & \leq & 9 \\
 & x_1 & & & & & & & \leq & 1 \\
 & & & x_2 & & & & & \leq & 1 \\
 & & & & & x_3 & & & \leq & 1 \\
 & & & & & & & x_4 & \leq & 1 \\
 & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}
$$

Example: a binary problem with one constraint (3/3)

- The ratios c_j/a_j are $3, 5/3, 4/3, 9/7$, in non-increasing order
- Dantzig's rule sets to 1 the variables in this order, as long as the constraint is satisfied, and gives to the first variable that would violate it the largest fractional value allowed: this gives an optimal solution of LP
- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP})$ is:

$$\tilde{x}_1 = \tilde{x}_2 = 1, \qquad \tilde{x}_3 = \frac{2}{3}, \qquad \tilde{x}_4 = 0$$

- The optimal value $z(\mathrm{LP})$ is

$$6 + 5 + 8(2/3) = 49/3$$

Example: the cover closure (1/2)

- The four-variable instance has the following seven covers:

C	$\sum_{j \in C} a_j$	$\lvert C \rvert$	Cover inequality	Minimal
$\{2,4\}$	10	2	$x_2 + x_4 \leq 1$	yes
$\{3,4\}$	13	2	$x_3 + x_4 \leq 1$	yes
$\{1,2,3\}$	11	3	$x_1 + x_2 + x_3 \leq 2$	yes
$\{1,2,4\}$	12	3	$x_1 + x_2 + x_4 \leq 2$	no
$\{1,3,4\}$	15	3	$x_1 + x_3 + x_4 \leq 2$	no
$\{2,3,4\}$	16	3	$x_2 + x_3 + x_4 \leq 2$	no
$\{1,2,3,4\}$	18	4	$x_1 + x_2 + x_3 + x_4 \leq 3$	no

Example: the cover closure (2/2)

- Only the first three covers are minimal; their inequalities already describe the cover closure
- Denote by $\mathrm{LP}^{\mathrm{cov}}$ the LP obtained by adding them to LP
- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}^{\mathrm{cov}})$ is:

$$\tilde{x}_1 = 1, \qquad \tilde{x}_2 = \frac{3}{5}, \qquad \tilde{x}_3 = \frac{2}{5}, \qquad \tilde{x}_4 = \frac{2}{5}$$

- The optimal value is

$$z(\mathrm{LP}^{\mathrm{cov}}) = 79/5 < z(\mathrm{LP}) = 49/3$$

Summary

Covers and cover inequalities:

- A cover C has $\sum_{j \in C} a_j > b$; its inequality $\sum_{j \in C} x_j \leq |C| - 1$ is valid for X
- The **minimal covers**, with the bounds $x_j \leq 1$, imply all the cover inequalities

Cover closure:

- $\text{conv}(X) \subseteq P^{\text{COV}} \subseteq P$; the covers can be exponentially many, so they are separated
- Four-variable instance: 7 covers, 3 minimal, $z(\text{LP}^{\text{COV}}) = 79/5 < z(\text{LP}) = 49/3$

Geometric interpretation

Introduction: Geometric interpretation

- With two or three variables we can **see** what a cover inequality cuts off
- We start from the domain alone, $[0,1]^2$ and $[0,1]^3$, then add a knapsack constraint
- The pictures show successive cuts, the cover closure and, in two instances, the convex hull $\text{conv}(X)$

How to read the figures

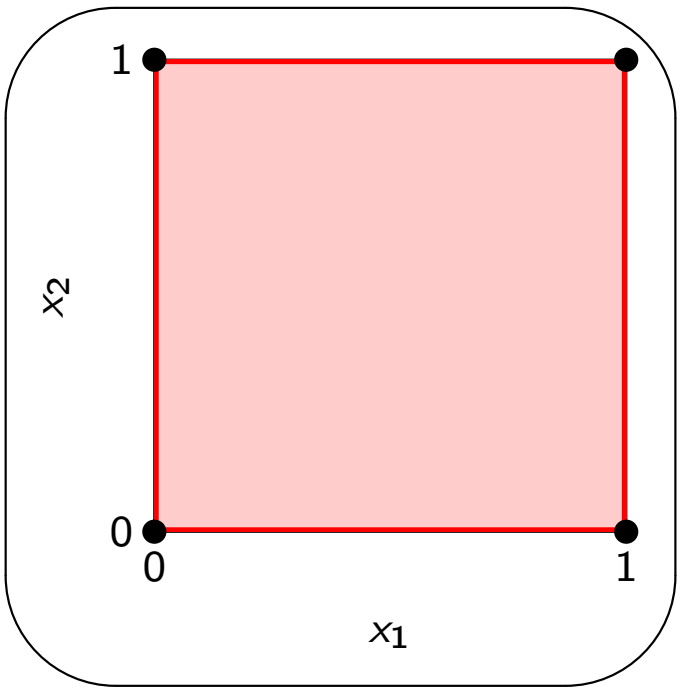
- All coordinates in the figures are the actual variable values
- The ticks mark the coordinates of the blue point and of the main vertices; a few vertices have no tick, to keep the axes readable
- In 3D, dashed edges are hidden behind the visible faces
- Every step is illustrated both in two and in three dimensions, with the same colours in all figures:
  - red: the continuous region under consideration
  - green: a cover inequality (an edge in 2D, a face in 3D)
  - black points: the feasible binary points
  - blue point: the fractional point discussed in the text

First consider only the domain, $[0,1]^2$ or $[0,1]^3$:

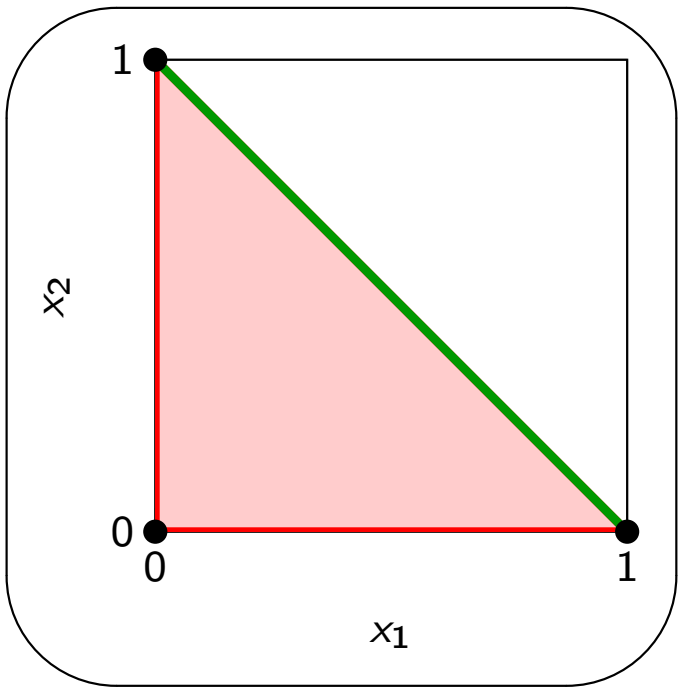
no constraint other than the bounds
is imposed in the next two examples.

Example: a cover inequality in the unit square

- In $[0,1]^2$, the inequality $x_1 + x_2 \leq 1$ removes only the binary point $(1,1)$ and preserves the binary points $(0,0)$, $(1,0)$ and $(0,1)$



Unit square: domain only.



Domain with $x_1 + x_2 \leq 1$.

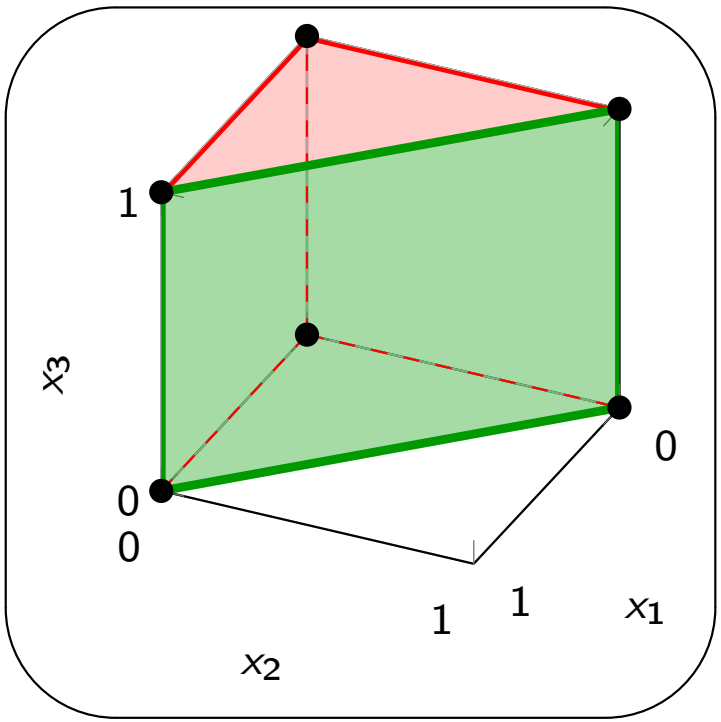
Example: cover inequalities in the unit cube (1/3)

- In $[0,1]^3$, the inequality $x_1 + x_2 \leq 1$ is independent of x_3; its boundary is parallel to the x_3 axis
- Similarly, the boundary of $x_1 + x_3 \leq 1$ is parallel to the x_2 axis
- The two halfspaces remove different portions of the cube
- Their intersection preserves exactly the binary points

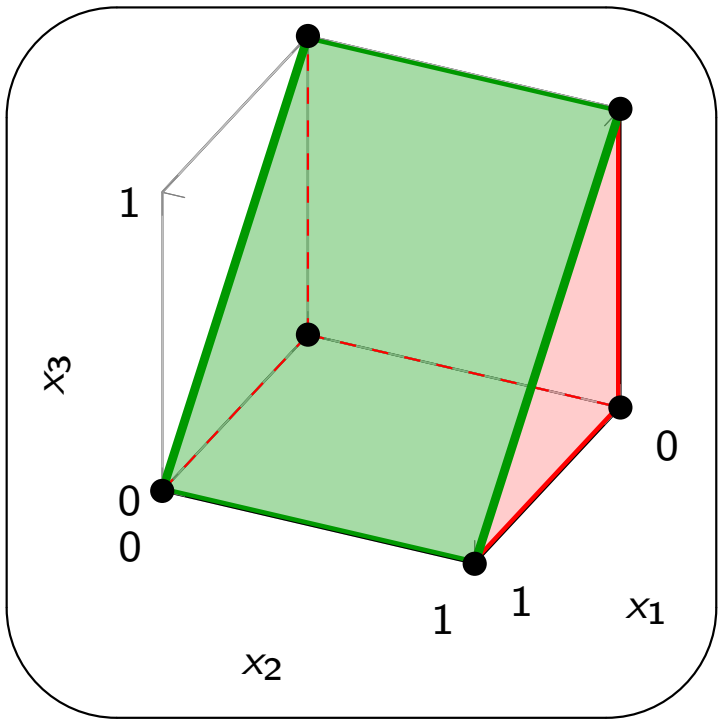
$$(0,0,0),\ (1,0,0),\ (0,1,0),\ (0,0,1),\ (0,1,1)$$

- The inequality $x_1 + x_2 + x_3 \leq 2$ removes only the binary point $x_1 = x_2 = x_3 = 1$

Example: cover inequalities in the unit cube (2/3)

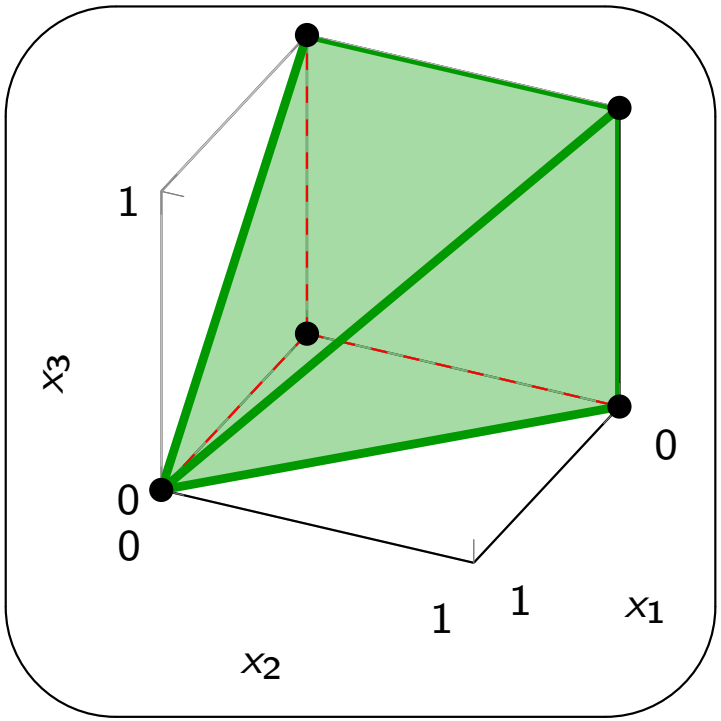


Domain with $x_1 + x_2 \leq 1$.

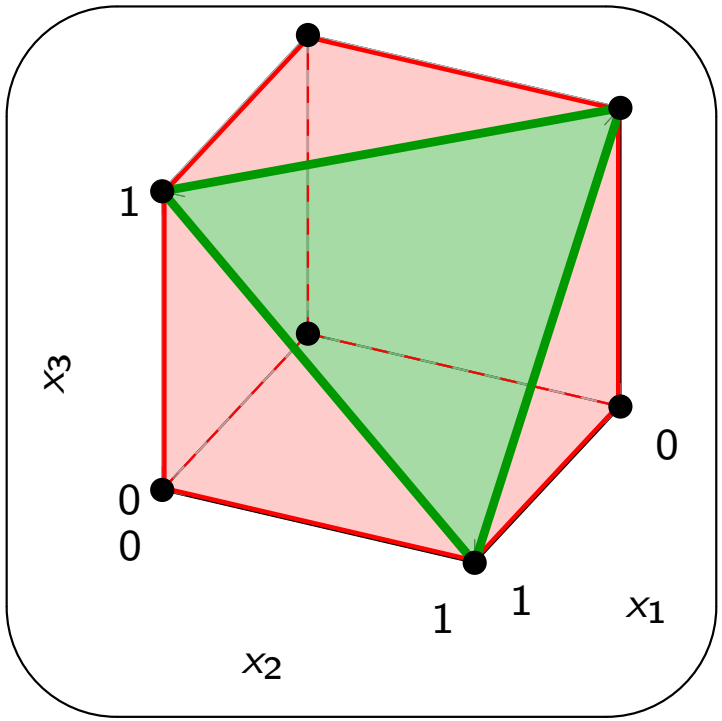


Domain with $x_1 + x_3 \leq 1$.

Example: cover inequalities in the unit cube (3/3)



Domain with both inequalities.



Domain with $x_1 + x_2 + x_3 \leq 2$.

Now we add a knapsack constraint to the domain:
first in two dimensions, then in three.

Example: a cover in two dimensions (1/2)

- For $\mathscr{I} = \{1, 2\}$, consider

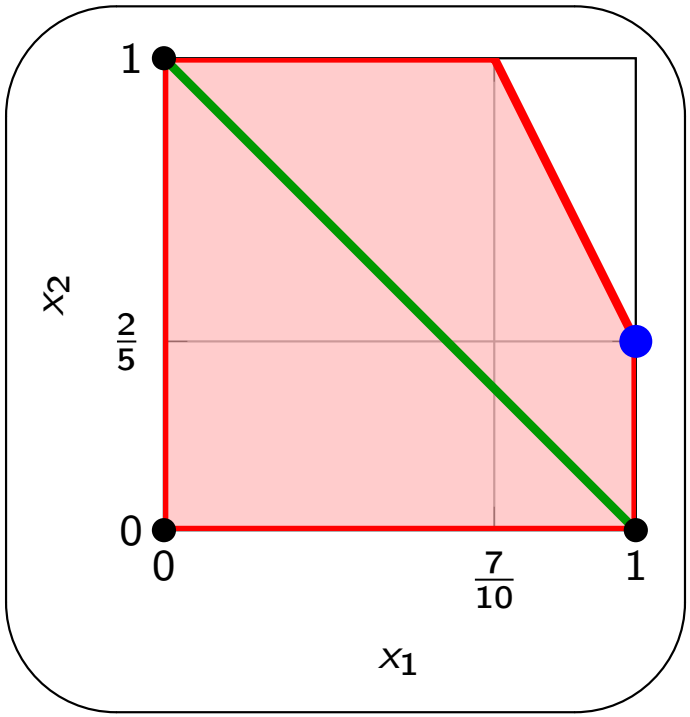
$$P_2 = \{\boldsymbol{x} \in [0, 1]^2 : 10x_1 + 5x_2 \leq 12\}$$

- The cover $C = \{1, 2\}$ gives $x_1 + x_2 \leq 1$
- The point $\tilde{x}_1 = 1$, $\tilde{x}_2 = 2/5$ satisfies the constraint at equality, but violates the cover by $2/5$
- The original polygon has vertices

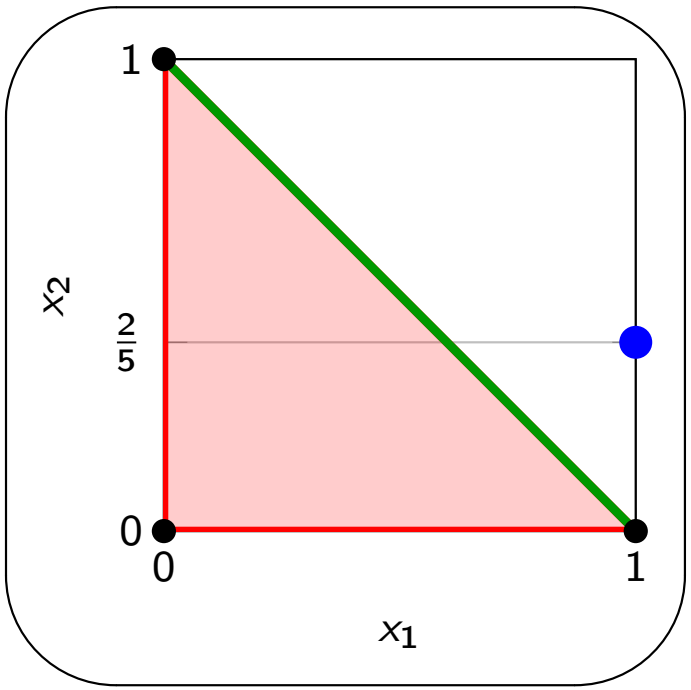
$$(0, 0),\ (1, 0),\ (1, 2/5),\ (7/10, 1),\ (0, 1)$$

- After adding the cut, the region is the triangle with vertices $(0, 0), (1, 0), (0, 1)$: all three feasible binary points are preserved, and the constraint becomes redundant, since $10x_1 + 5x_2 \leq 10 < 12$ on the triangle

Example: a cover in two dimensions (2/2)



Original P_2 and the boundary $x_1 + x_2 = 1$.



$P_2 \cap \{x_1 + x_2 \leq 1\}$: the blue point is removed.

Example: successive cover cuts (1/5)

- For $\mathscr{I} = \{1, 2, 3\}$, let

$$P_A = \{\boldsymbol{x} \in [0, 1]^3 : 10x_1 + 5x_2 + 6x_3 \leq 12\}$$

- There are exactly three covers:

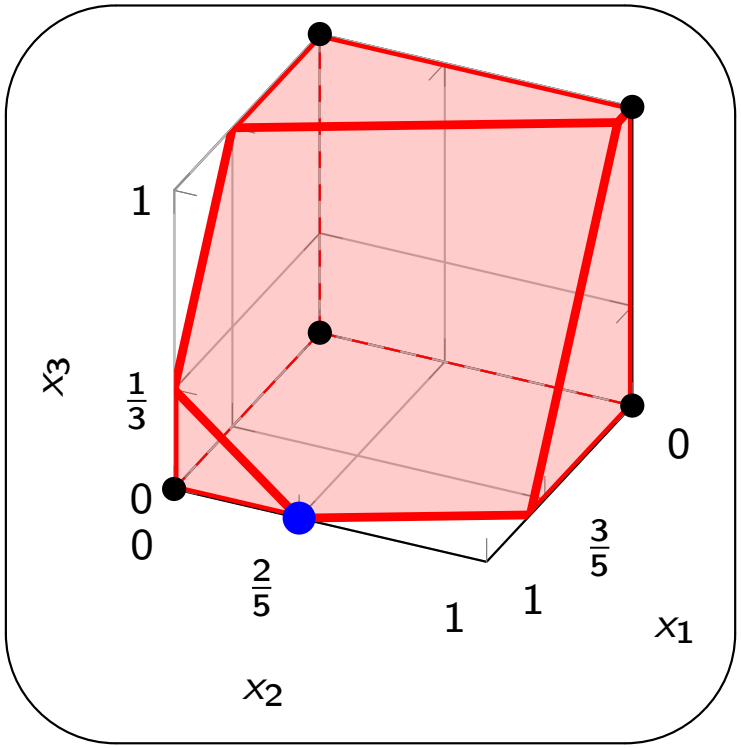
$$\begin{array}{rcl} \{1, 2, 3\} & : & x_1 + x_2 + x_3 \leq 2, \\ \{1, 3\} & : & x_1 + x_3 \leq 1, \\ \{1, 2\} & : & x_1 + x_2 \leq 1. \end{array}$$

- Only the last two are minimal

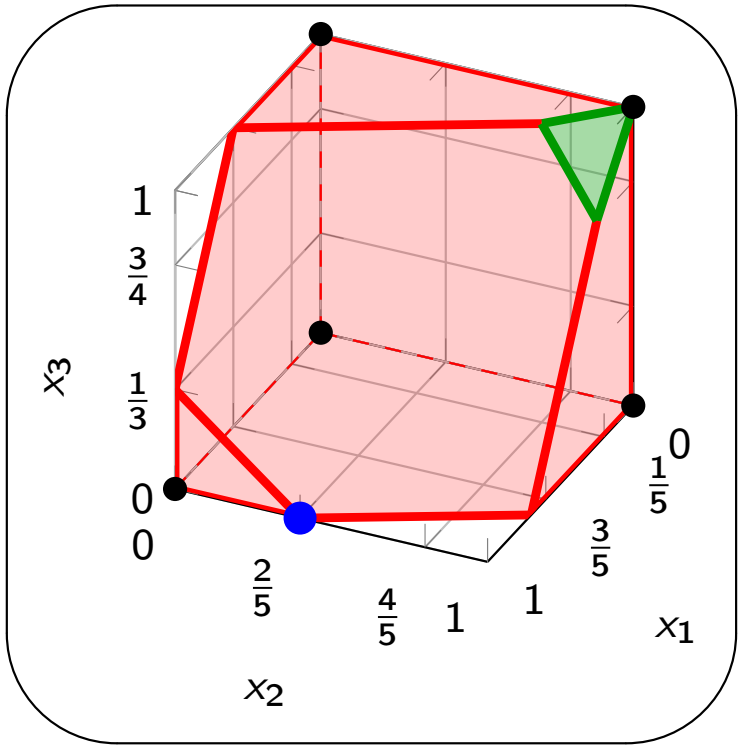
Example: successive cover cuts (2/5)

- The blue point $\tilde{x}_1 = 1$, $\tilde{x}_2 = 2/5$, $\tilde{x}_3 = 0$ lies in P_A
- It satisfies the first inequality and is on the boundary of the second
- The third is violated by 2/5, exactly as in the two-dimensional example
- Unlike the previous domain-only panels, every following panel includes the constraint $10x_1 + 5x_2 + 6x_3 \leq 12$

Example: successive cover cuts (3/5)

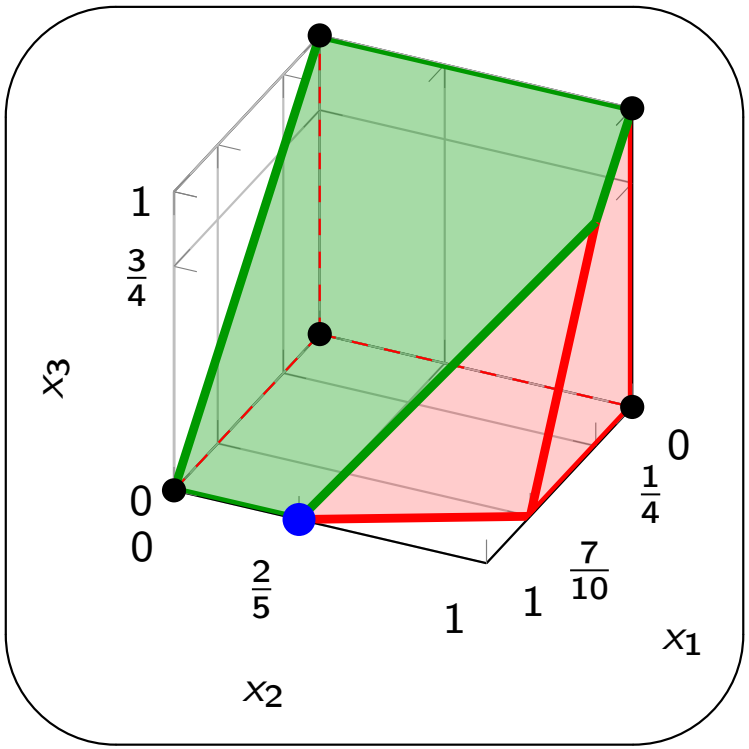


Original P_A.

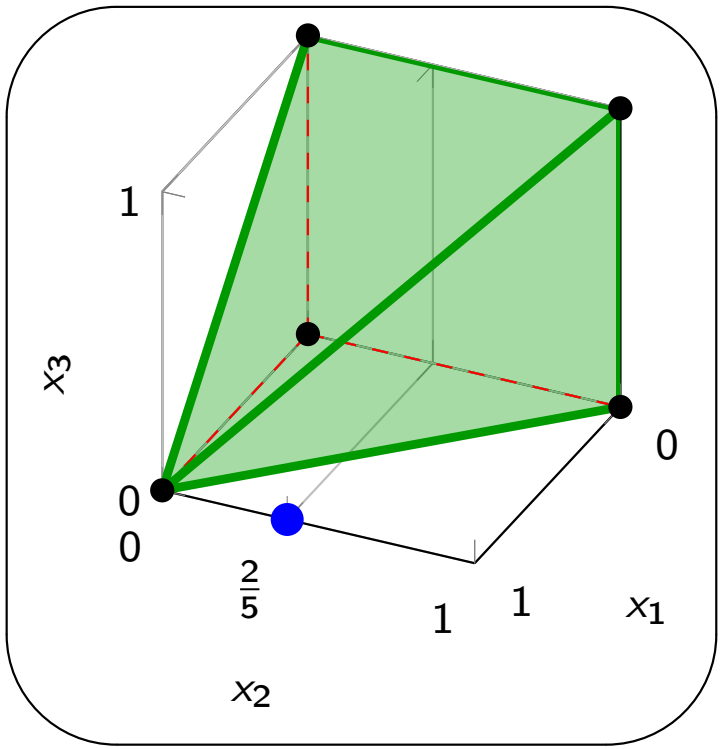


Add $x_1 + x_2 + x_3 \leq 2$.

Example: successive cover cuts (4/5)



Also add $x_1 + x_3 \leq 1$.



Also add $x_1 + x_2 \leq 1$: the cover closure.

Example: successive cover cuts (5/5)

- For the objective max $40x_1 + 15x_2 + 12x_3$, the optimal solution over P_A is $\tilde{\boldsymbol{x}} = (1, 2/5, 0)$, with value 46
- Over the cover closure, the optimal solution is $(1, 0, 0)$, with value 40, which is also optimal for the binary problem
- In this instance the cover closure coincides with $\text{conv}(X)$: its vertices, shown in the last panel, are all binary
- This is a property of this instance, not of every cover closure

Example: a different constraint (1/3)

- For $\mathscr{I} = \{1, 2, 3\}$, consider now the instance

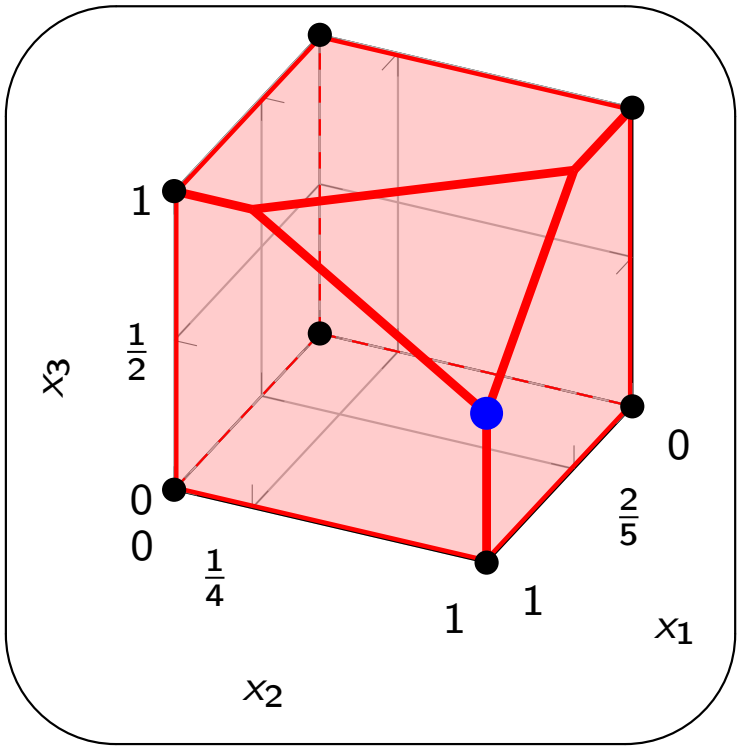
$$P_B = \{\boldsymbol{x} \in [0, 1]^3 : 5x_1 + 4x_2 + 6x_3 \leq 12\}$$

- Every pair of variables satisfies the constraint, whereas the three coefficients sum to $15 > 12$
- Thus the only cover is $\{1, 2, 3\}$ and it is minimal; its inequality is

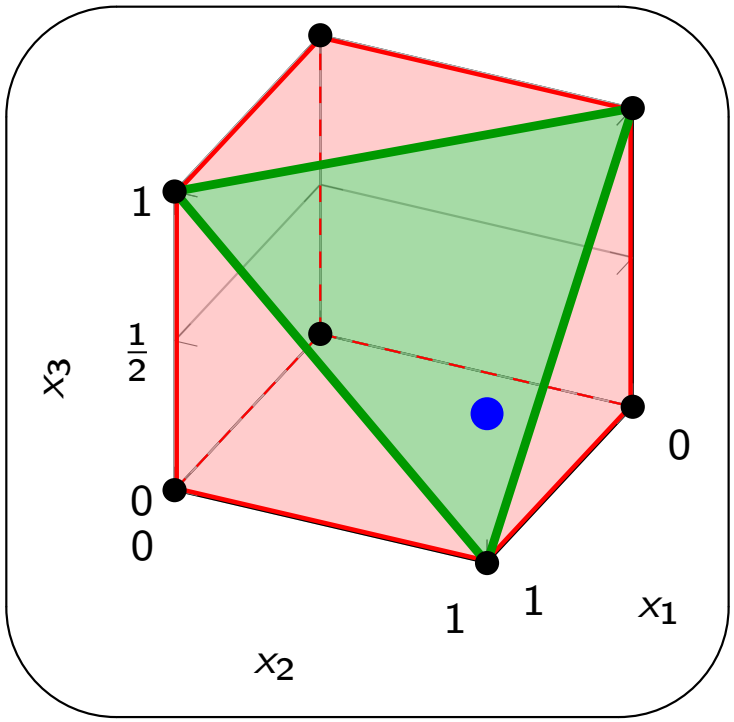
$$x_1 + x_2 + x_3 \leq 2$$

- The blue point $\tilde{x}_1 = \tilde{x}_2 = 1$, $\tilde{x}_3 = 1/2$ has left-hand side 12 and violates this cut by $1/2$

Example: a different constraint (2/3)



Original P_B.



$P_B \cap \{x_1 + x_2 + x_3 \leq 2\}$.

Example: a different constraint (3/3)

- The plane of the constraint intersects the cube edges at $(1, 1, 1/2)$, $(1, 1/4, 1)$ and $(2/5, 1, 1)$
- The cover removes these fractional vertices while preserving all seven feasible binary points
- As in the 2D example, after the cut the constraint is redundant: the sum of the two largest coefficients is $6 + 5 = 11 < 12$
- The polyhedron $P_B \cap \{x_1 + x_2 + x_3 \leq 2\}$ has only binary vertices (last panel): here a single cover inequality gives $\text{conv}(X)$

Summary

- In $[0,1]^2$ and $[0,1]^3$ a cover inequality removes only the binary points with all the variables of the cover equal to 1
- With a knapsack constraint, a cover inequality cuts off **fractional vertices** and keeps all the feasible binary points; after the cut the constraint can become redundant
- In P_A the cover closure is $\text{conv}(X)$, and in P_B a single cover inequality gives it: properties of these instances, not general facts

Separation of cover inequalities

Introduction: Separation of cover inequalities

- The covers can be exponentially many: we add a cover inequality only when a point violates it
- We define the **cover separation problem** and write it as an ILP problem with one constraint
- On the four-variable instance we separate the cover inequalities iteratively, until none is violated

The cover separation problem

- Given a point $\tilde{\boldsymbol{x}} \in P$, the **cover separation problem** (CSP) consists in finding a cover inequality violated by $\tilde{\boldsymbol{x}}$, or in proving that $\tilde{\boldsymbol{x}}$ satisfies all the cover inequalities

- Formally, the CSP asks to find a cover

$$C \in \mathcal{C}: \qquad \sum_{j \in C} \tilde{x}_j > |C| - 1,$$

or to prove that

$$\sum_{j \in C} \tilde{x}_j \leq |C| - 1 \qquad \forall\, C \in \mathcal{C}.$$

When is a cover inequality violated?

- Typically, $\tilde{\boldsymbol{x}}$ is an optimal solution of the current LP relaxation, but any point of P can be separated
- For a cover C, the cover inequality is violated by $\tilde{\boldsymbol{x}}$ if

$$\sum_{j \in C} \tilde{x}_j > |C| - 1$$

- Moving $\sum_{j \in C} \tilde{x}_j$ to the right-hand side and 1 to the left-hand side, and writing $|C| = \sum_{j \in C} 1$, we obtain equivalent conditions (next slide)

From the violation to a cost

- The equivalent conditions are:

$$|C| - \sum_{j \in C} \tilde{x}_j < 1,$$

$$\sum_{j \in C} 1 - \sum_{j \in C} \tilde{x}_j < 1,$$

$$\sum_{j \in C} (1 - \tilde{x}_j) < 1.$$

- Assigning to each index j the cost $q_j = 1 - \tilde{x}_j \geq 0$, the CSP amounts to finding a cover of **minimum cost**

A minimum-cost cover

A violated cover inequality exists
if and only if
the minimum cost $\sum_{j \in C}(1 - \tilde{x}_j)$ of a cover
is **less than** 1.

An ILP formulation for the CSP

- Introduce n binary variables z_j: $z_j = 1$ if j belongs to the cover and $z_j = 0$ otherwise
- The objective minimizes the total cover cost
- The single constraint forces the selected set to be a cover

The model $\mathrm{ILP}_{\mathrm{CSP}}$

$$(\mathrm{ILP}_{\mathrm{CSP}}) \qquad \min \quad \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j) z_j$$

$$\text{subject to} \quad \sum_{j \in \mathscr{I}} a_j z_j \geq b + 1,$$

$$z_j \in \{0, 1\}, \quad \forall j \in \mathscr{I}.$$

- The variables z_j select the indices of the cover
- The constraint $\sum_{j \in \mathscr{I}} a_j z_j \geq b + 1$ says that the selected set is a cover

Separation and the value of $\mathrm{ILP}_{\mathrm{CSP}}$

Proposition 3

A cover inequality violated by $\tilde{\boldsymbol{x}}$ *exists if and only if* $z(\mathrm{ILP}_{\mathrm{CSP}}) < 1$.

Proof

- Every feasible assignment of the binary variables z_j represents a cover $C = \{j : z_j = 1\}$, and its objective is

$$\sum_{j \in C} (1 - \tilde{x}_j)$$

- By the equivalent conditions above, this is less than 1 exactly when the corresponding cut is violated
- Taking the minimum over all covers proves both implications
- The CSP is feasible because $\mathscr{I}$ itself is a cover

□

The maximally violated cover inequality

- Let $\tilde{z}_1, \tilde{z}_2, \ldots, \tilde{z}_n$ be an optimal solution of $\mathrm{ILP}_{\mathrm{CSP}}$
- If $z(\mathrm{ILP}_{\mathrm{CSP}}) < 1$, the cover

$$\tilde{C} = \left\{ j \in \mathscr{I} : \ \tilde{z}_j = 1 \right\}$$

  is associated with the cover inequality $\sum_{j \in \tilde{C}} x_j \leq |\tilde{C}| - 1$ **maximally violated** by $\tilde{\boldsymbol{x}}$
- Otherwise, no cover inequality is violated by $\tilde{\boldsymbol{x}}$

Optimal and feasible solutions of $\mathrm{ILP}_{\mathrm{CSP}}$

- To find a violated cover inequality, it is not necessary to solve $\mathrm{ILP}_{\mathrm{CSP}}$ to optimality: any feasible solution $\bar{z}_1, \ldots, \bar{z}_n$ of cost less than 1 gives the cover $\bar{C} = \{j \in \mathscr{I} : \bar{z}_j = 1\}$, whose cover inequality is violated by $\tilde{\boldsymbol{x}}$
- Conversely, a feasible solution of cost at least 1 does **not** prove that no violated cover inequality exists: this requires the optimal value (or a lower bound) of $\mathrm{ILP}_{\mathrm{CSP}}$
- If $z(\mathrm{ILP}_{\mathrm{CSP}}) < 1$, the maximum violation of a cover inequality at $\tilde{\boldsymbol{x}}$ is $1 - z(\mathrm{ILP}_{\mathrm{CSP}})$

Example: separation of the cover inequalities (1/10)

- We consider again the four-variable instance
- The cover inequalities are separated **iteratively**: at each iteration,
  - the current LP is solved
  - the $\mathrm{ILP}_{\mathrm{CSP}}$ is solved for its optimal solution $\tilde{\boldsymbol{x}}$
  - the maximally violated cover inequality is added

  until no cover inequality is violated

Example: separation of the cover inequalities (2/10)

Iteration 1. An optimal solution of LP is

$$\tilde{x}_1 = 1, \qquad \tilde{x}_2 = 1, \qquad \tilde{x}_3 = \frac{2}{3}, \qquad \tilde{x}_4 = 0, \qquad z(\mathrm{LP}) = \frac{49}{3}.$$

The associated $\mathrm{ILP}_{\mathrm{CSP}}$ is:

$$\begin{array}{lrcrcrcrcl} \min & & & & & \frac{1}{3}\, z_3 & + & z_4 & & \\ \text{subject to} & 2\, z_1 & + & 3\, z_2 & + & 6\, z_3 & + & 7\, z_4 & \geq & 10 \\ & z_1, & & z_2, & & z_3, & & z_4 & \in & \{0,1\} \end{array}$$

Example: separation of the cover inequalities (3/10)

Iteration 1.

- An optimal solution of $\mathrm{ILP}_{\mathrm{CSP}}$ is $\tilde{z}_1 = \tilde{z}_2 = \tilde{z}_3 = 1$, $\tilde{z}_4 = 0$, with value $z(\mathrm{ILP}_{\mathrm{CSP}}) = \frac{1}{3} < 1$

- Accordingly, $\tilde{C} = \{j \in \mathscr{J} : \tilde{z}_j = 1\} = \{1, 2, 3\}$ is the cover associated with the cover inequality maximally violated by $\tilde{\boldsymbol{x}}$:

$$x_1 + x_2 + x_3 \leq 2, \qquad \text{since} \qquad \tilde{x}_1 + \tilde{x}_2 + \tilde{x}_3 = 1 + 1 + \frac{2}{3} = \frac{8}{3} > 2.$$

Example: separation of the cover inequalities (4/10)

Iteration 2. Adding this cover inequality to LP, we obtain the following LP, denoted LP_1:

$$
\begin{array}{rrcrcrcrcl}
\max & 6\,x_1 & + & 5\,x_2 & + & 8\,x_3 & + & 9\,x_4 & & \\
\text{subject to} & 2\,x_1 & + & 3\,x_2 & + & 6\,x_3 & + & 7\,x_4 & \leq & 9 \\
& x_1 & + & x_2 & + & x_3 & & & \leq & 2 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}
$$

Example: separation of the cover inequalities (5/10)

Iteration 2. An optimal solution of LP_1 is

$$\tilde{x}_1 = 1, \qquad \tilde{x}_2 = 1, \qquad \tilde{x}_3 = 0, \qquad \tilde{x}_4 = \frac{4}{7}, \qquad z(\mathrm{LP}_1) = \frac{113}{7}.$$

The associated $\mathrm{ILP}_{\mathrm{CSP}}$ is:

$$\begin{array}{lrcrcrcrcl} \min & & & & & z_3 & + & \frac{3}{7} z_4 & & \\ \text{subject to} & 2 z_1 & + & 3 z_2 & + & 6 z_3 & + & 7 z_4 & \geq & 10 \\ & z_1, & & z_2, & & z_3, & & z_4 & \in & \{0,1\} \end{array}$$

Example: separation of the cover inequalities (6/10)

Iteration 2.

- An optimal solution of $\mathrm{ILP}_{\mathrm{CSP}}$ is $\tilde{z}_1 = 0$, $\tilde{z}_2 = 1$, $\tilde{z}_3 = 0$, $\tilde{z}_4 = 1$, with value $z(\mathrm{ILP}_{\mathrm{CSP}}) = \frac{3}{7} < 1$
- The cover $\{1, 2, 4\}$ has the same cost; we choose the minimal cover
- Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{z}_j = 1\} = \{2, 4\}$ and the maximally violated cover inequality is

$$x_2 + x_4 \leq 1, \qquad \text{since} \qquad \tilde{x}_2 + \tilde{x}_4 = 1 + \frac{4}{7} = \frac{11}{7} > 1.$$

Example: separation of the cover inequalities (7/10)

Iteration 3. Adding this cover inequality to LP_1, we obtain the following LP, denoted LP_2:

$$
\begin{array}{rrcrcrcrcl}
\max & 6\,x_1 & + & 5\,x_2 & + & 8\,x_3 & + & 9\,x_4 & & \\
\text{subject to} & 2\,x_1 & + & 3\,x_2 & + & 6\,x_3 & + & 7\,x_4 & \leq & 9 \\
& x_1 & + & x_2 & + & x_3 & & & \leq & 2 \\
& & & x_2 & & & + & x_4 & \leq & 1 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}
$$

Example: separation of the cover inequalities (8/10)

Iteration 3. An optimal solution of LP_2 is

$$\tilde{x}_1 = 1, \qquad \tilde{x}_2 = \frac{3}{5}, \qquad \tilde{x}_3 = \frac{2}{5}, \qquad \tilde{x}_4 = \frac{2}{5}, \qquad z(\mathrm{LP}_2) = \frac{79}{5}.$$

The associated $\mathrm{ILP}_{\mathrm{CSP}}$ is:

$$\begin{array}{rrcrcrcrcl} \min & & & \frac{2}{5}\, z_2 & + & \frac{3}{5}\, z_3 & + & \frac{3}{5}\, z_4 & & \\ \text{subject to} & 2\, z_1 & + & 3\, z_2 & + & 6\, z_3 & + & 7\, z_4 & \geq & 10 \\ & z_1, & & z_2, & & z_3, & & z_4 & \in & \{0,1\} \end{array}$$

Example: separation of the cover inequalities (9/10)

Iteration 3.

- An optimal solution of $\mathrm{ILP}_{\mathrm{CSP}}$ is $\tilde{z}_1 = 0$, $\tilde{z}_2 = 1$, $\tilde{z}_3 = 0$, $\tilde{z}_4 = 1$, with value $z(\mathrm{ILP}_{\mathrm{CSP}}) = 1$
- The covers $\{1, 2, 3\}$ and $\{1, 2, 4\}$ have the same cost

No cover inequality is violated by $\tilde{x}$,
and the separation stops.

Example: separation of the cover inequalities (10/10)

- The final value $z(\mathrm{LP}_2) = \frac{79}{5}$ coincides with the value $z(\mathrm{LP}^{\mathrm{cov}})$ of the cover closure, computed in the example of the cover closure
- The separation reaches the bound of the cover closure by adding only 2 of the 7 cover inequalities
- The solution $\tilde{\boldsymbol{x}}$ is still **fractional**, although it satisfies all the cover inequalities
- The cover closure does not coincide, in general, with the convex hull of the integer solutions

Summary

- The **CSP** finds a cover inequality violated by $\tilde{\boldsymbol{x}}$, or proves that none exists
- A cover inequality is violated if and only if $\sum_{j \in C}(1 - \tilde{x}_j) < 1$: we look for a cover of **minimum cost**, with the model $\mathrm{ILP}_{\mathrm{CSP}}$
- A violated cover inequality exists if and only if $z(\mathrm{ILP}_{\mathrm{CSP}}) < 1$; the maximum violation is $1 - z(\mathrm{ILP}_{\mathrm{CSP}})$
- Four-variable instance: 2 separated covers give $z(\mathrm{LP}_2) = z(\mathrm{LP}^{\mathrm{cov}}) = 79/5$, with a fractional $\tilde{\boldsymbol{x}}$

Transforming the CSP into a complementary problem

Introduction: The complementary problem

- Complementing the variables of the CSP gives a 0–1 problem with one $\leq$ constraint and a maximization objective: the **complementary problem**
- Its feasible solutions are in one-to-one correspondence with the covers
- It suggests a simple **greedy heuristic** for the separation, which we apply to the four-variable instance

Complementary variables

- Consider a feasible assignment of the CSP variables z_j
- Define the complementary binary variables y_j by

$$y_j = 1 - z_j \qquad \forall j \in \mathscr{J}$$

- Thus $y_j = 1$ if j is outside the cover, whereas $y_j = 0$ if j belongs to the cover

The constraint and the objective

- The constraint becomes

$$\sum_{j \in \mathscr{I}} a_j(1 - y_j) \geq b + 1 \quad \Longleftrightarrow \quad \sum_{j \in \mathscr{I}} a_j y_j \leq \sum_{j \in \mathscr{I}} a_j - (b + 1)$$

  By the assumption $\sum_{j \in \mathscr{I}} a_j > b$, this right-hand side is non-negative

- The objective can be written as

$$\sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)(1 - y_j) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j) y_j$$

- Since the first two terms are constant, minimizing it is equivalent to the problem of the next slide, called the **complementary problem**, with non-negative (possibly zero) objective coefficients $1 - \tilde{x}_j$

The complementary problem $\mathrm{ILP}^2_{\mathrm{CSP}}$

$$(\mathrm{ILP}^2_{\mathrm{CSP}}) \qquad \max \quad \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j) y_j$$

$$\text{subject to} \quad \sum_{j \in \mathscr{I}} a_j y_j \leq \sum_{j \in \mathscr{I}} a_j - (b+1),$$

$$y_j \in \{0,1\}, \qquad \forall j \in \mathscr{I}.$$

- The variables y_j select the indices **outside** the cover
- Complementation is a bijection between feasible solutions, so

$$z(\mathrm{ILP}_{\mathrm{CSP}}) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}})$$

Reading the cover from $\mathrm{ILP}^2_{\mathrm{CSP}}$

- Let $\tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_n$ be an optimal solution of $\mathrm{ILP}^2_{\mathrm{CSP}}$
- If $n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) < 1$, the cover

$$\tilde{C} = \left\{ j \in \mathscr{I} : \ \tilde{y}_j = 0 \right\}$$

  is associated with the cover inequality $\sum_{j \in \tilde{C}} x_j \leq |\tilde{C}| - 1$ **maximally violated** by $\tilde{\boldsymbol{x}}$
- Otherwise, no cover inequality is violated by $\tilde{\boldsymbol{x}}$

Feasible solutions of $\mathrm{ILP}^2_{\mathrm{CSP}}$

- Any feasible solution $\bar{y}_1, \ldots, \bar{y}_n$ of $\mathrm{ILP}^2_{\mathrm{CSP}}$ gives the cover

$$\bar{C} = \{j \in \mathscr{I} : \ \bar{y}_j = 0\},$$

with cost

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\bar{y}_j$$

- If this cost is less than 1, the cover inequality of $\bar{C}$ is violated by $\tilde{\boldsymbol{x}}$

Feasible solutions of the complementary problem are easy to build:

a **greedy** procedure gives one quickly.

A greedy separation heuristic

The following greedy procedure constructs feasible values $\bar{y}_j$ for the complementary problem.

1. Set $\bar{y}_j = 0$ for all $j \in \mathscr{J}$ and the residual right-hand side $r = \sum_{j \in \mathscr{J}} a_j - (b+1)$. Sort the indices by non-increasing ratio $q_j/a_j = (1 - \tilde{x}_j)/a_j$, breaking ties by increasing index.

2. Scan the indices in this order. If $q_j > 0$ and $a_j \leq r$, set $\bar{y}_j = 1$ and $r := r - a_j$; otherwise skip it.

3. Reconstruct the cover $C = \{j : \bar{y}_j = 0\}$ and evaluate its cost. If desired, reduce it to a minimal cover by removing indices as long as $\sum_{j \in C} a_j$ exceeds b.

Properties of the greedy heuristic

- Indices with $q_j = 0$ ($\tilde{x}_j = 1$) are never selected: they remain in the cover, which may then be non-minimal
- Removing an index from a cover decreases its cost by $q_j \geq 0$: reducing a violated cover to a minimal one cannot decrease its violation
- The greedy algorithm is a **heuristic** for the complementary problem: if it does not find a violated cover inequality, we cannot conclude that none exists

Example: complementary problem and greedy scan (1/4)

- Consider the point $\tilde{\boldsymbol{x}} = (1, 1, \frac{2}{3}, 0)$ of Iteration 1 in the separation example
- The two constants of the complementary problem are

$$4 - \sum_{j \in \mathscr{J}} \tilde{x}_j = 4 - 1 - 1 - \frac{2}{3} = \frac{4}{3},$$

$$\sum_{j \in \mathscr{J}} a_j - (b + 1) = 2 + 3 + 6 + 7 - (9 + 1) = 8.$$

Example: complementary problem and greedy scan (2/4)

- The complementary problem is

$$\begin{array}{lrcrcrcrcl} \max & & & & & \frac{1}{3}\,y_3 & + & y_4 & & \\ \text{subject to} & 2\,y_1 & + & 3\,y_2 & + & 6\,y_3 & + & 7\,y_4 & \leq & 8 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

- The costs are $q_j = 1 - \tilde{x}_j$: $q_1 = q_2 = 0$, $q_3 = \frac{1}{3}$, $q_4 = 1$

Example: complementary problem and greedy scan (3/4)

- The greedy scan, with the residual right-hand side r starting from 8:

j	q_j	a_j	q_j/a_j	Decision	Residual
4	1	7	$\frac{1}{7}$	$\bar{y}_j = 1$	1
3	$\frac{1}{3}$	6	$\frac{1}{18}$	$\bar{y}_j = 0$	1
1	0	2	0	$q_j = 0$, $\bar{y}_j = 0$	1
2	0	3	0	$q_j = 0$, $\bar{y}_j = 0$	1

- The scan gives $\bar{y}_1 = \bar{y}_2 = \bar{y}_3 = 0$ and $\bar{y}_4 = 1$, of value 1

Example: complementary problem and greedy scan (4/4)

- Indices 3 and 4 cannot be taken together; the only alternatives with positive objective coefficient have values 1/3 and 1
- Thus this greedy solution is also optimal, and

$$z(\mathrm{ILP}_{\mathrm{CSP}}) = 4 - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) = \frac{4}{3} - 1 = \frac{1}{3}$$

- The cover is $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{1, 2, 3\}$, as found in Iteration 1 of the separation example

Summary

- With $y_j = 1 - z_j$ the CSP becomes the **complementary problem** $\mathrm{ILP}^2_{\mathrm{CSP}}$: maximize $\sum_{j \in \mathscr{J}} (1 - \tilde{x}_j) y_j$ with one $\leq$ constraint
- $z(\mathrm{ILP}_{\mathrm{CSP}}) = n - \sum_{j \in \mathscr{J}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}})$, and the cover is $\{j : \tilde{y}_j = 0\}$
- The **greedy heuristic** scans the indices by non-increasing q_j / a_j: it can find a violated cover inequality, but its failure proves nothing
- On the four-variable instance the greedy solution is optimal and gives the cover $\{1, 2, 3\}$

Extended cover inequalities

Introduction: Extended cover inequalities

- A cover inequality involves only the variables of the cover
- Variables outside the cover with large coefficients can be added to its left-hand side, with the same right-hand side
- The result is a stronger valid inequality: on the four-variable instance it closes the gap

The largest coefficient of a cover

- A cover inequality can be strengthened by also considering the variables outside the cover whose coefficients are at least as large as every coefficient of the cover
- Let

$$\widehat{a}(C) = \max_{j \in C} a_j$$

be the maximum coefficient of the variables in C

Extended cover inequalities

Definition 6: Extended cover inequality

For a cover C, let

$$E(C) = C \cup \{j \in \mathscr{I} \setminus C : a_j \geq \hat{a}(C)\}.$$

The associated extended cover inequality is

$$\sum_{j \in E(C)} x_j \leq |C| - 1.$$

The right-hand side depends on $|C|$, not on $|E(C)|$.

Validity of extended cover inequalities

Proposition 4

For every cover $C \in \mathcal{C}$, the extended cover inequality

$$\sum_{j \in E(C)} x_j \leq |C| - 1$$

is valid for X.

Proof

- Let $k = |C|$ and take any set $S \subseteq E(C)$ with $|S| = k$
- The sets $S \setminus C$ and $C \setminus S$ have the same number of variables
- Each coefficient a_j with $j \in S \setminus C$ is at least $\widehat{a}(C)$, i.e., at least as large as each coefficient with $j \in C \setminus S$
- Replacing the indices of $C \setminus S$ with those of $S \setminus C$ gives

$$\sum_{j \in S} a_j \geq \sum_{j \in C} a_j > b$$

- Hence no point $\boldsymbol{x} \in X$ can have k variables of $E(C)$ equal to 1: at most $k - 1$ of them can be equal to 1 together

□

Dominance and lifting

- Since $C \subseteq E(C)$ and the right-hand side is the same, the extended cover inequality **dominates** the cover inequality of C
- It can be seen as a **lifting** of the coefficients of the variables in $E(C) \setminus C$ from 0 to 1
- It does not forbid a variable with a large coefficient to be equal to 1 together with variables of C: it limits to $|C| - 1$ the number of variables of the whole set $E(C)$ equal to 1

Example: strengthening a cover (1/3)

- In the four-variable instance, consider the cover $C = \{1, 2, 3\}$
- Then $\widehat{a}(C) = 6$, and the coefficient of x_4 is $7 \geq 6$, so

$$E(C) = \{1, 2, 3, 4\}, \qquad x_1 + x_2 + x_3 + x_4 \leq 2$$

Example: strengthening a cover (2/3)

- The strengthened LP, denoted $\mathrm{LP}_{\mathrm{ext}}$, is

$$
\begin{array}{rrcrcrcrcl}
\max & 6\,x_1 & + & 5\,x_2 & + & 8\,x_3 & + & 9\,x_4 & & \\
\text{subject to} & 2\,x_1 & + & 3\,x_2 & + & 6\,x_3 & + & 7\,x_4 & \leq & 9 \\
& x_1 & + & x_2 & + & x_3 & + & x_4 & \leq & 2 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}
$$

Example: strengthening a cover (3/3)

- An optimal solution of $\mathrm{LP}_{\mathrm{ext}}$ is $\tilde{\boldsymbol{x}} = (1, 0, 0, 1)$, with value

$$z(\mathrm{LP}_{\mathrm{ext}}) = 6 + 9 = 15$$

- Since it is binary, $\tilde{\boldsymbol{x}}$ is also optimal for the binary problem

- We have

$$z(\mathrm{LP}_{\mathrm{ext}}) = 15 < z(\mathrm{LP}^{\mathrm{cov}}) = 79/5$$

- In this instance, one extended cover inequality is stronger than all the cover inequalities together

Summary

- $E(C)$ adds to C the indices outside C with $a_j \geq \widehat{a}(C) = \max_{j \in C} a_j$
- The **extended cover inequality** $\sum_{j \in E(C)} x_j \leq |C| - 1$ is valid and dominates the cover inequality of C: the right-hand side stays $|C| - 1$
- Four-variable instance: $x_1 + x_2 + x_3 + x_4 \leq 2$ gives $z(\mathrm{LP}_{\mathrm{ext}}) = 15$ with the binary optimal solution $(1, 0, 0, 1)$

Violation and distance from a cover inequality

Introduction: Violation and distance

- How much does a point violate a cover inequality, and how far is it from it?
- We define the violation, the Euclidean distance from the boundary hyperplane and the orthogonal projection
- The two measures can rank the covers differently: the CSP maximizes the violation, not the distance

The boundary hyperplane

- Let $C \subseteq \mathscr{I}$ be a cover. Its cover inequality is

$$\sum_{j \in C} x_j \leq |C| - 1,$$

and its boundary hyperplane is

$$H_C = \left\{ \boldsymbol{x} \in \mathbb{R}^n : \sum_{j \in C} x_j = |C| - 1 \right\}$$

Violation

- The violation of the cover inequality at a point $\tilde{\boldsymbol{x}}$ is

$$v_C(\tilde{\boldsymbol{x}}) = \max \left\{ 0, \sum_{j \in C} \tilde{x}_j - (|C| - 1) \right\}$$

- The point violates the cover inequality precisely when $v_C(\tilde{\boldsymbol{x}}) > 0$

Distance

- The coefficients of the cover inequality are 1 for $j \in C$ and 0 otherwise, so their Euclidean norm is $\sqrt{|C|}$
- Therefore, the Euclidean distance of $\tilde{\boldsymbol{x}}$ from H_C is

$$d(\tilde{\boldsymbol{x}}, H_C) = \frac{\left|\sum_{j \in C} \tilde{x}_j - (|C| - 1)\right|}{\sqrt{|C|}}$$

- If the cover inequality is violated,

$$d(\tilde{\boldsymbol{x}}, H_C) = \frac{v_C(\tilde{\boldsymbol{x}})}{\sqrt{|C|}}$$

Orthogonal projection

- The orthogonal projection $\boldsymbol{x}^{\perp}$ of $\tilde{\boldsymbol{x}}$ onto H_C has components

$$x_j^{\perp} = \begin{cases} \tilde{x}_j - \dfrac{\sum_{k \in C} \tilde{x}_k - (|C| - 1)}{|C|}, & j \in C, \\ \tilde{x}_j, & j \in \mathscr{I} \setminus C. \end{cases}$$

- Indeed, substitution gives $\sum_{j \in C} x_j^{\perp} = |C| - 1$

Violation is not distance

- The CSP maximizes the **violation**, not the **distance**: when the covers have different cardinalities, the two criteria can rank them differently
- For example, at the point $\tilde{\boldsymbol{x}} = (1/10, 1, 1)$ of P_A (the example of successive cover cuts), the covers $\{1, 2\}$, $\{1, 3\}$ and $\{1, 2, 3\}$ all have violation $1/10$
- Their distances are $1/(10\sqrt{2})$, $1/(10\sqrt{2})$ and $1/(10\sqrt{3})$, respectively

Example: orthogonal projection in two dimensions (1/2)

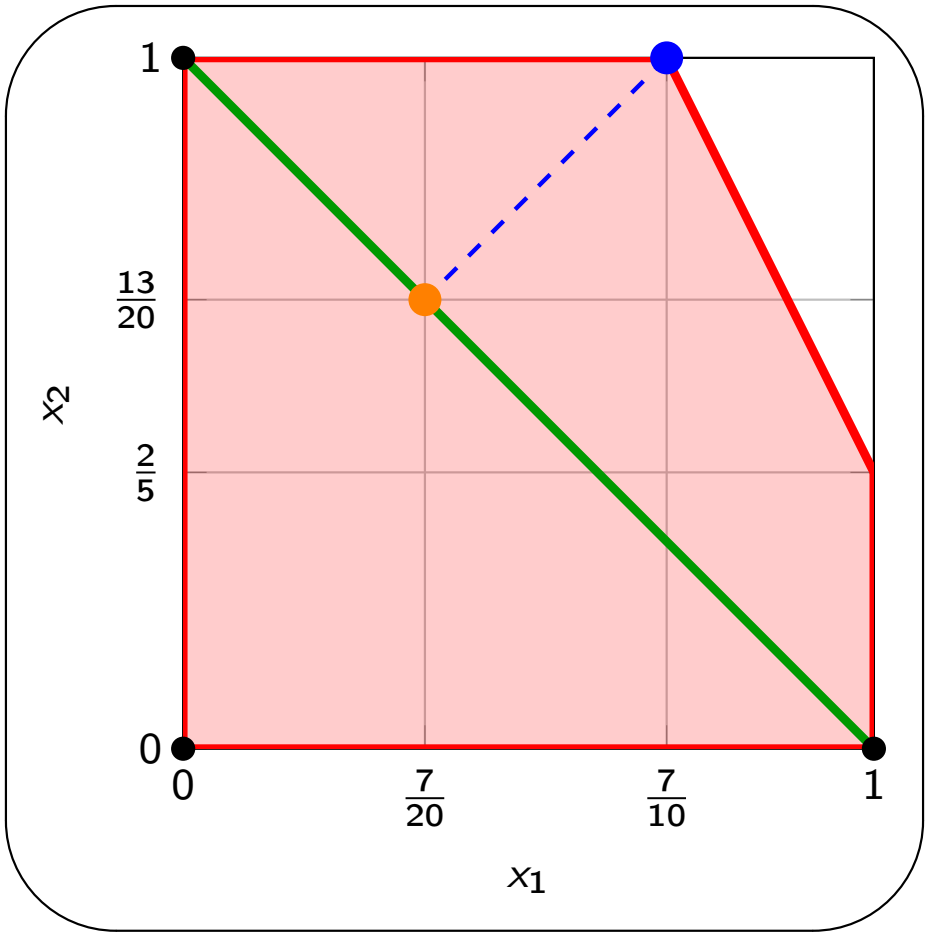
- Take the vertex $\tilde{x}_1 = 7/10$, $\tilde{x}_2 = 1$ of P_2 (the example of a cover in two dimensions) and the cover $C = \{1, 2\}$
- Its cover inequality is $x_1 + x_2 \leq 1$ and $|C| = 2$. Then

$$v_C(\tilde{\boldsymbol{x}}) = \frac{7}{10}, \qquad x_1^{\perp} = \frac{7}{20}, \qquad x_2^{\perp} = \frac{13}{20},$$

$$d(\tilde{\boldsymbol{x}}, H_C) = \frac{7}{10\sqrt{2}}$$

- In the figure on the next slide, the red region is P_2 and the green edge is the boundary H_C
- The blue dashed segment is perpendicular to H_C and the orange point is the orthogonal projection

Example: orthogonal projection in two dimensions (2/2)



Example: orthogonal projection in three dimensions (1/2)

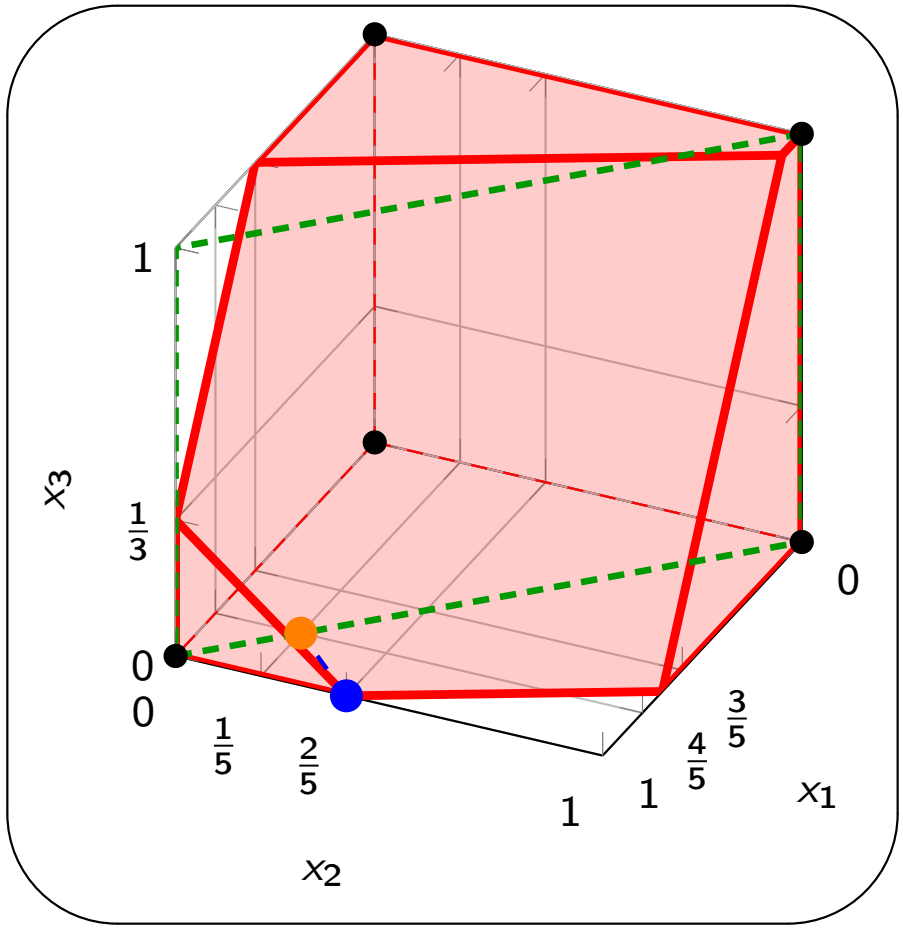
- Take the point $\tilde{x}_1 = 1$, $\tilde{x}_2 = 2/5$, $\tilde{x}_3 = 0$ of P_A (the example of successive cover cuts) and the cover $C = \{1, 2\}$
- Its cover inequality is $x_1 + x_2 \leq 1$ and $|C| = 2$. Then

$$v_C(\tilde{\boldsymbol{x}}) = \frac{2}{5}, \qquad x_1^{\perp} = \frac{4}{5}, \qquad x_2^{\perp} = \frac{1}{5}, \qquad x_3^{\perp} = 0,$$

$$d(\tilde{\boldsymbol{x}}, H_C) = \frac{2}{5\sqrt{2}}$$

- In the figure on the next slide, the red region is P_A and the green dashed rectangle is H_C restricted to the unit cube
- Since $3 \notin C$, the projection does not change $\tilde{x}_3$

Example: orthogonal projection in three dimensions (2/2)



Summary

- Violation: $v_C(\tilde{\boldsymbol{x}}) = \max\{0, \sum_{j \in C} \tilde{x}_j - (|C| - 1)\}$
- Distance from H_C: $|\sum_{j \in C} \tilde{x}_j - (|C| - 1)| / \sqrt{|C|}$, that is $v_C(\tilde{\boldsymbol{x}}) / \sqrt{|C|}$ for a violated inequality
- The projection moves only the components in C, each by the same amount
- The CSP maximizes the **violation**: covers of different cardinalities with the same violation have different distances

Typical mistakes

Introduction: Typical mistakes

- We list the mistakes that are made most often with cover inequalities
- They concern the definition of a cover, the extended covers, the separation and the cover closure
- Knowing them helps to read and solve the exercises correctly

Typical mistakes

Observation 1: Typical mistakes (1/3)

- ***Using a set that is not a cover.*** *The inequality* $\sum_{j \in C} x_j \leq |C| - 1$ *is valid only if the total coefficient* $\sum_{j \in C} a_j$ *exceeds the right-hand side* b*; check the coefficients before writing it.*
- ***Using*** $|E(C)| - 1$ ***for an extended cover.*** *The right-hand side of the extended cover inequality is* $|C| - 1$*, where* C *is the cover, not* $|E(C)| - 1$*.*

Typical mistakes

Observation: Typical mistakes (2/3)

- ***Removing all the indices with $q_j = 0$ from a cover.*** *To obtain a minimal cover, remove indices one at a time as long as the remaining set is still a cover: removing all of them may leave a set that is not a cover.*
- ***Stopping the separation at a heuristic failure.*** *If the greedy heuristic algorithm finds no violated cover inequality, we cannot conclude that none exists: only the optimal value of the separation problem proves it.*

Typical mistakes

Observation: Typical mistakes (3/3)

- ***Confusing violation and distance.*** *The cover separation problem maximizes the violation; covers with the same violation can have different distances from the point.*
- ***Believing that the cover closure is the convex hull.*** *A point can satisfy all the cover inequalities and still be fractional (the separation example).*

Summary

Typical mistakes:

- Check that $\sum_{j \in C} a_j > b$; the right-hand side of an extended cover is $|C| - 1$
- Reduce a cover to a minimal one index by index; a heuristic failure proves nothing
- Violation is not distance; the cover closure is not the convex hull

In summary

Introduction: In summary

- We collect the key ideas of the chapter
- We close with the dashboard of the four-variable instance: the bounds found in this chapter

Summary (1/3)

- A cover is a set of variables whose total coefficient exceeds the right-hand side; its cover inequality allows at most $|C| - 1$ of its variables to be equal to 1, and it is valid for X
- The minimal covers are enough: their inequalities imply all the others, but they can be exponentially many, so they are separated only when violated

Summary (2/3)

- Separating a cover inequality means finding a cover of minimum cost $\sum_{j \in C}(1 - \tilde{x}_j)$: a violated inequality exists if and only if the minimum cost is less than 1; this is a 0–1 problem with one constraint, defined in the chapter
- An extended cover inequality dominates the cover inequality of the same cover
- The cover closure is closer to conv(X) than the LP relaxation, but in general it is not equal to it

Summary (3/3)

Dashboard of the four-variable instance.

- The lower and upper bounds on the optimal value of the binary problem; the objective coefficients are integer, so the upper bound is the rounded-down value of the LP relaxation

Source	Lower bound	Upper bound	From
LP relaxation	11	16	greedy solution $(1, 1, 0, 0)$, $\lfloor z(\mathrm{LP}) \rfloor$ with $z(\mathrm{LP}) = \frac{49}{3}$
Cover closure	11	15	$\lfloor z(\mathrm{LP}^{\mathrm{cov}}) \rfloor$ with $z(\mathrm{LP}^{\mathrm{cov}}) = \frac{79}{5}$, 2 separated covers
Extended cover	15	15	$z(\mathrm{LP}_{\mathrm{ext}}) = 15$, optimal solution $(1, 0, 0, 1)$